Martha Fey

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RESEARCH INTERESTS

Environmental health; statistical analysis; water quality; environmental epidemiology; human health outcomes

EDUCATION

Bachelor of Science in Sustainability Science

Aug 2021 - May 2025

University of Iowa - Iowa City, Iowa

GPA: 3.90/4.0

Academic Standing: Top 5% of graduating class

RESEARCH EXPERIENCE

Research Assistant

Mar 2023 - June 2025

Ecoclimatology of Carbon and Water (ECH2O) Lab -
University of Iowa

- Independently constructed an Eddy Covariance Flux Tower to collect continuous ecosystem-atmosphere exchange data.
- Integrated solar power, battery storage, and remote data transmission systems for autonomous environmental monitoring.
- Collected and analyzed environmental datasets to investigate ecosystem processes and climate-related dynamics.
- Contributed to research examining the biogeophysical cooling potential of restored prairie ecosystems using Eddy Covariance and thermal UAS remote sensing.

PUBLICATIONS

Pacer, R., Meerdink, S., Swanson, B., Ramker, N., Fey, M., et al. (2026). Jun 2020 – Feb 2022
Assessing the Biogeophysical Cooling Potential of Constructed Prairie with Eddy Covariance & Thermal UAS Remote Sensing.
Manuscript under review.

ENVIRONMENTAL HEALTH EXPERIENCE

Environmental Health and Land Use Assistant

Mar 2026 – Present

Benton County - Vinton, Iowa

- Applied ArcGIS Pro and reviewed historical records to develop spatial datasets and maps supporting environmental health and land-use evaluation processes.
- Conducted water sample collection and laboratory preparation for over 65 private wells.
- Explained water test results and recommended systems to homeowners based on results of water tests.
- Cleaned and organized water quality data for R and developed a proposed research approach examining relationships between well characteristics, including age, depth, and water quality results.

TECHNICAL SKILLS

- ArcGIS Pro
- R
- Python